Systems, Drawn · Video #6

AI SRE Agents on 3 Real Incidents: Where They'd Help, Where They'd Hurt

Diagram pack · CC BY 4.0 · SRE // microservices

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Sources: AWS post-event summary (Oct 2025) · GitHub RCA (17 Aug 2026) and availability report (Aug 2026) · Google Cloud Service Health, incident mSFVt6PiP557bjRyYsut (Oct 2026) · vendor surveys: New Relic, Catchpoint, PagerDuty, Grafana Labs · vendor claims: PagerDuty, Datadog, Cognition (Devin). Each diagram cites its own source. AWS and GCS times in PDT (UTC−7) as reported, ART = PDT + 4 h; GitHub in UTC.

1. d0a_opening_pager

B01

11:48 PM PDT · 19 Oct 2025 · AWS us-east-1

03:48 ART · 20 Oct

AI agent (hypothetical)

dashboard errors ↑

illustrative



on-call human

Investigate → would help

Recover → STUCK

"no established operational recovery procedure"

— AWS post-event summary (EC2 lease system)

help / hurt = our analysis · source: aws.amazon.com/message/101925 · ART = PDT + 4 h

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2. d0b_opening_three_incidents

B01

AWS us-east-1
Oct 2025

would help

would hurt

GitHub
17 Aug 2026

would help

would hurt

Google Cloud Storage
us-central1 · Oct 2026

would help

would hurt

our analysis · no agent was used

Should it hold the pager?

three real incidents, each from the company's own report · no agent was on call in any of them

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3. d2a_aws_dns_race

B02 · D2

AWS us-east-1 · 19–20 Oct 2025

automation, not AI · our framing

DynamoDB DNS management (simplified, per AWS)

DNS Enactor A (delayed)

its one check at start went STALE → wrote an OLD plan over a newer one

regional endpoint DNS record: EMPTY

dynamodb.us-east-1 .amazonaws.com

DNS Enactor B

applied the newest plan; its clean-up deleted that old plan

"a latent race condition in the DynamoDB DNS management system that resulted in an incorrect empty DNS record" — AWS post-event summary

would help · trace errors → one empty DNS record

DNS restored 2:25 AM PDT · 20 Oct

06:25 ART

our analysis · AWS describes automated DNS management and names no AI · two of AWS's three DNS Enactors (one per AZ)
source: AWS post-event summary, aws.amazon.com/message/101925 · times as AWS reports them (PDT) · ART = PDT + 4 h

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4. d2b_aws_recovery

AWS us-east-1 · 20 Oct 2025 · after DNS came back

DNS restored 2:25 AM PDT · 20 Oct

06:25 ART · increased errors on new EC2 launches continued (per AWS)

would hurt · EC2 lease recovery: no established procedure

"Since this situation had no established operational recovery procedure…"
— AWS, about DWFM (the EC2 lease system)
our analysis: an agent that follows runbooks has nothing to follow

event ended
2:20 PM PDT · 20 Oct

18:20 ART

"We have already disabled the DynamoDB DNS Planner and the DNS Enactor automation worldwide."

— AWS post-event summary

help/hurt = our analysis · EC2 normal 1:50 PM PDT (17:50 ART); whole event 11:48 PM PDT 19 Oct → 2:20 PM PDT 20 Oct (per AWS)
= 03:48 ART 20 Oct → 18:20 ART 20 Oct
source: AWS post-event summary, aws.amazon.com/message/101925 · times as AWS reports them (PDT) · ART = PDT + 4 h

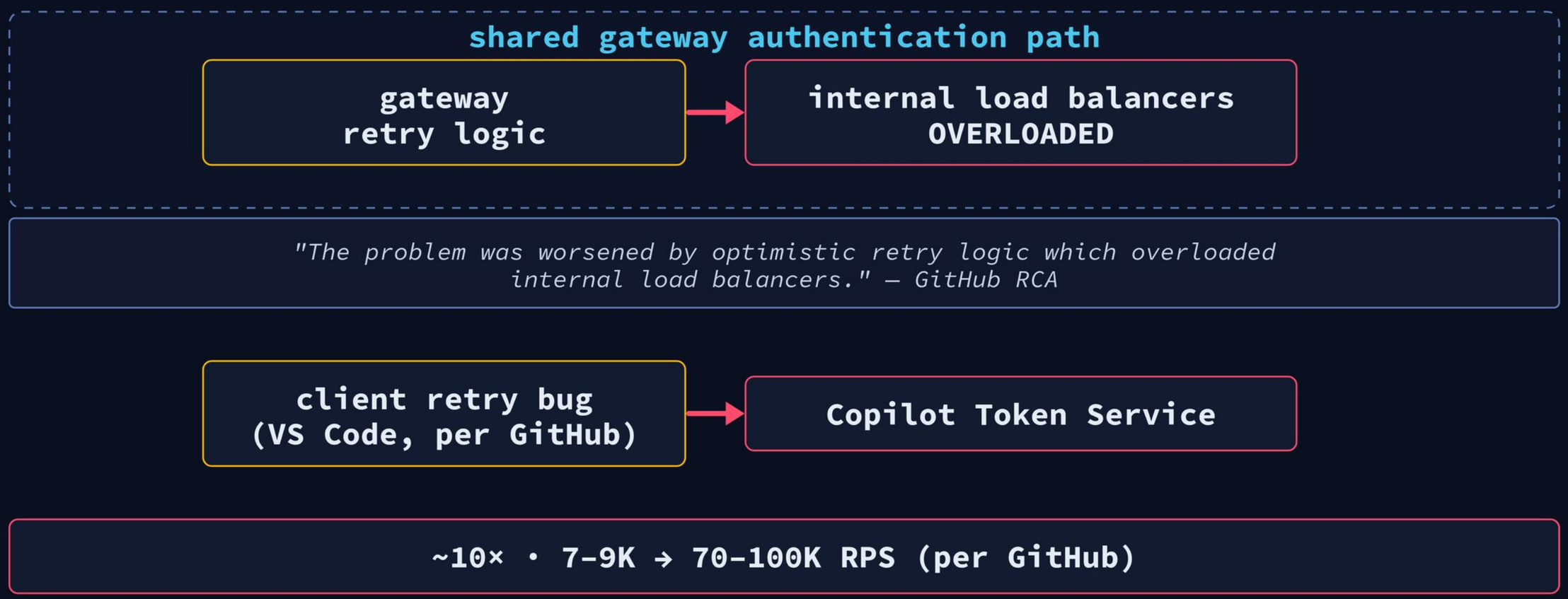
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5. d3a_github_retry_layers

B03 · D3

GitHub · 17 Aug 2026 · per GitHub's RCA

two retry layers · our framing



simplified · GitHub doesn't publish this topology · 'two retry layers' = our framing
(GitHub: "Review retry and backoff limits across gateways and clients") · incident 13:28–21:15 UTC (10:28–18:15 ART)
sources: githubstatus.com/incidents/zkxwbgr0cnmx (RCA) · github.blog availability report, Aug 2026

Systems, Drawn · CC BY 4.0 · src/d3a_github_retry_layers.d2

6. d3b_github_block_403

B03 · D3

GitHub · 17 Aug 2026 · where an agent would help, where it would hurt

would help
spot the ~10× jump

would hurt · suggested step

adds load to an overloaded path

loop

retry / restart

blocked incoming token requests with a 403 (per GitHub)

then ramped traffic back gradually

per GitHub's RCA: the 403 was applied at the load balancers in Northern VA (the overload in d3a was Central US)
help/hurt = our analysis · the 403 was one of several measures (with reducing gateway retries and pausing HAProxy)
source: githubstatus.com/incidents/zkxwbgr0cnmx (RCA, posted 17 Aug 2026 21:15 UTC)

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7. d4a_gcs_capacity_blocked

B04 · D4

12h30m of write errors · us-central1 · 8 Oct 2026 · per Google

Add emergency capacity: BLOCKED

"blocked by systemic regional capacity limits" — Google

detected 04:50 PDT → first status-page update 11:24 PDT
= 6h34m (our arithmetic)

08:50 ART → 15:24 ART

would help · your own alerts, early

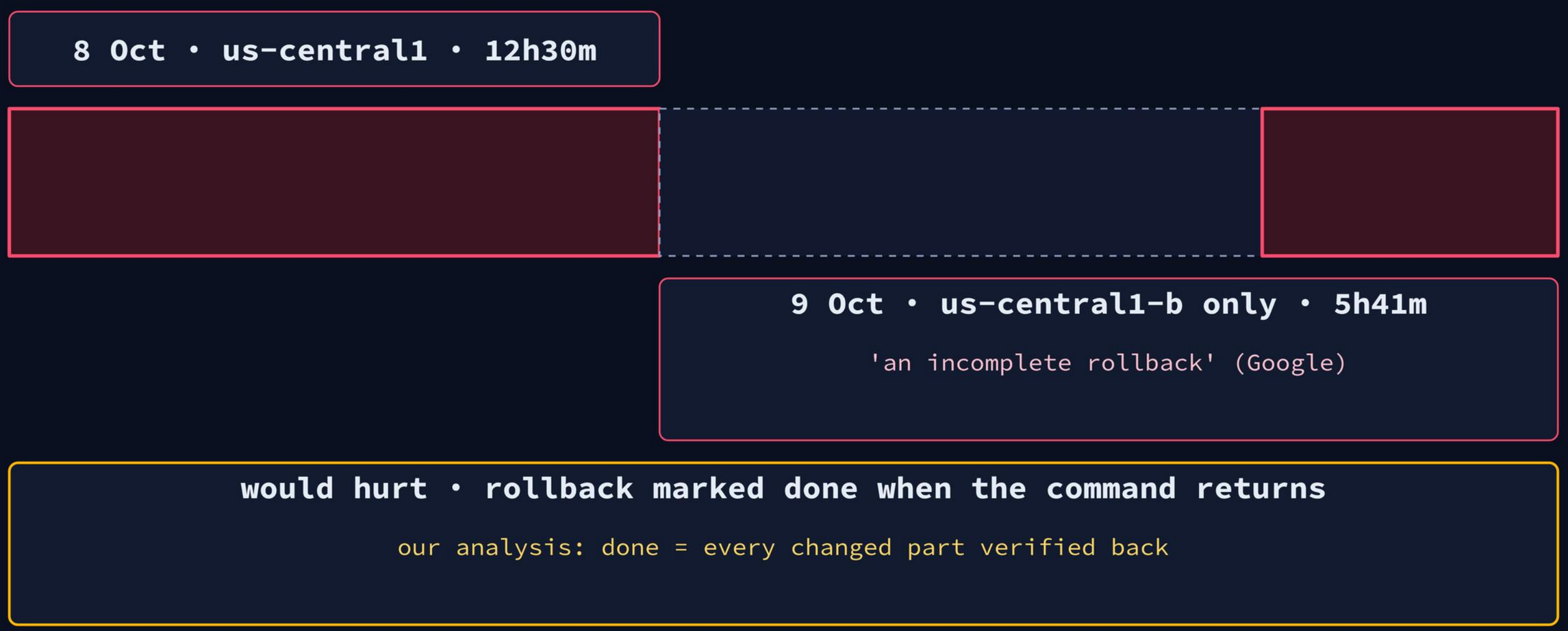
Google: Service disruption

detection time from Google's final report (14 Oct) · our arithmetic · 11:24 PDT = first entry on the status page
help = our analysis · 'Service disruption' = Google's label · times PDT (UTC-7), ART = PDT + 4 h
source: Google Cloud Service Health, status.cloud.google.com/incidents/mSFVt6PiP557bjRyYsut

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8. d4b_gcs_two_windows

B04 · D4



to scale, 60 px = 1 h · grey = 11 h 35 min between the windows (our arithmetic), not an all-clear
8 Oct 04:00–16:30 PDT (08:00–20:30 ART), 'Incident End' (Google: "temporary relief")
9 Oct 04:05–09:46 PDT (08:05–13:46 ART), zone us-central1-b only
source: Google incident report, 14 Oct 2026 · status.cloud.google.com/incidents/mSFVt6PiP557bjRyYsut

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9. d5_pattern_summary

B05 · D5

our analysis · no agent was used

AWS us-east-1 Oct 2025	GitHub 17 Aug 2026	Google Cloud Storage us-central1 · Oct 2026

Investigate: would help

Act · the risky step was a decision

turn automation off	block requests → users see errors	verify the rollback

our analysis of each company's own report (AWS post-event summary · GitHub RCA · Google incident report)
green = would help · amber = the risky step is a decision

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10. d1_ai_sre_loop

B06 · D1

the shape both vendors describe · our summary



every box proposes · none acts

Investigation: AI's highest operational value (35.8%)

New Relic 2026 Observability Forecast · vendor survey (with ETR) · n=2,575 respondents · Apr–May 2026 · question not published

boxes = our summary of PagerDuty's and Datadog's own product descriptions (vendor claims, not measurements)
purple = vendor description · sources: PagerDuty SRE Agent GA (30 Oct 2025) · Datadog Bits AI SRE press release (2 Dec 2025)
New Relic 2026 Observability Forecast, PDF p.7 (35.8%), p.2 and p.25 (n=2,575)

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11. d1b_vendor_claims

B06

Then why do the vendors say it works?

"classifies incidents, surfaces critical context from past incidents, and recommends remediation steps"

— PagerDuty (SRE Agent GA, 30 Oct 2025) · vendor claim

"investigates alerts and surfaces actionable root cause in minutes"

— Datadog (Bits AI SRE GA, 2 Dec 2025) · vendor claim

Cognition: Devin Oncall 100% on a 17-fault benchmark (own evaluation)
CloudThinker also 100% (public leaderboard)

vendor claim, not production

vendor claims, quoted · not measurements · sources: pagerduty.com (SRE Agent GA) · Datadog press release, 2 Dec 2025
devin.ai/blog/introducing-devin-oncall (SREGym-Lite, 17 faults; leaderboard as of 15 Oct 2026)

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12. d6a_survey_newrelic

B07

Only 6.1% of respondents operate without AI incident tools

New Relic 2026 Observability Forecast · vendor survey (with ETR) ·
n=2,575 respondents · Apr–May 2026 · question not published

About 1 in 4 respondents run AI agents in production with no monitoring

New Relic 2026 · vendor survey · n=2,575

vendor surveys of their own respondents

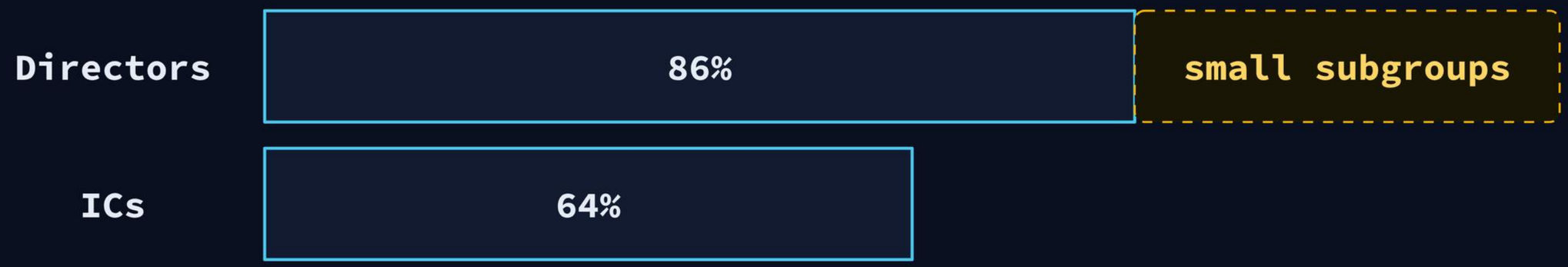
6.1%: PDF p.7 · 1 in 4 = 24.6% of all respondents, p.6 · p.25: all data from the 2,575-respondent survey
source: New Relic 2026 Observability Forecast, newrelic.com/sites/default/files/2026-09/new-relic-2026-observability-forecast-report.pdf

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13. d6b_survey_catchpoint

B07

at least somewhat comfortable with AI suggestions
during incident response



Catchpoint SRE Report 2026 · vendor survey · 418 responses · Jul–Aug 2025
subgroup sizes not published

to scale: 10 px = 1 point · 'comfortable' = somewhat + very comfortable · the chart excludes executives
amber dashed = the record disagrees or has a gap · source: Catchpoint, The SRE Report 2026, p.36 printed / p.37 PDF

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14. d6c_survey_pagerduty25

81% of execs say they trust AI agents to take action during a crisis, such as a service outage or security event

85% say their orgs need better procedures to detect AI errors

PagerDuty AI Resilience Survey 2025 · vendor · 1,500 Director+ execs
half IT / half business · Jul–Aug 2025

same survey, same 1,500 Director+ executives
PagerDuty's wording: "on the company's behalf" · "errors or failures in AI tools"
source: pagerduty.com/newsroom/ai-resilience-survey-2025 (23 Sep 2025) + method page (Wakefield Research)

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15. d6d_survey_b08

B08

44% require human involvement when AI remediates customer-facing systems

PagerDuty 2026 State of AI-First Operations · vendor
1,000 business and IT leaders and developers

77% think autonomous AI action is valuable highest skepticism by a wide margin

Grafana Labs Observability Survey 2026 · vendor · 1,363 responses · Oct 2025–Jan 2026

Act = a decision · our analysis

- **turn automation off**
- **block requests → users see errors**
- **verify the rollback**

Grafana: 'valuable' = somewhat valuable + very valuable + critical (its note 2)
two different vendor surveys, different questions and samples
sources: pagerduty.com 2026 State of AI-First Operations (17 Mar 2026) · grafana.com/observability-survey/2026

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16. d7_b13_no_independent_data

B13



amber dashed = the record disagrees or has a gap · absence of public data, not a claim about any tool

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